

STATA GROUP ASSIGNMENT 2

CAPM and Operating Profitability

Value-weighted deciles | July 1963–July 2026 | 757 months

1 Data and excess returns

We use ten value-weighted operating-profitability (OP) decile portfolios from Kenneth French's Data Library. Decile 1 has the lowest OP and decile 10 the highest. The source files were downloaded on September 15, 2026 (July 2026 CRSP vintage). Monthly market excess returns and risk-free returns come from the Fama/French three-factor file.

For each decile, excess return is $R_{d,t} - R_{f,t}$. We match the portfolios and factors by month, giving 757 observations per decile. Returns are in percent per month. The market variable, $\text{Mkt} - \text{RF}$, is already an excess return, so we do not subtract RF again.

2 CAPM regressions

For each decile, we estimate by ordinary least squares:

$$R_{d,t} - R_{f,t} = \alpha_d + \beta_d(R_{m,t} - R_{f,t}) + \varepsilon_{d,t}.$$

Alpha is the intercept, in percentage points per month. Beta measures sensitivity to the market excess return. We use the usual OLS standard errors, as reported by Stata's `regress` command.

Table 1. CAPM estimates with conventional OLS standard errors

Decile	α	$\text{SE}(\alpha)$	t_α	β	$\text{SE}(\beta)$	t_β	R^2
1	-0.3414	0.1144	-2.9844	1.3030	0.0254	51.2345	0.7766
2	-0.1437	0.0810	-1.7742	1.0980	0.0180	60.9881	0.8313
3	-0.1475	0.0653	-2.2592	1.0298	0.0145	70.9674	0.8696
4	-0.0196	0.0609	-0.3228	0.9706	0.0135	71.7284	0.8720
5	0.0904	0.0641	1.4091	0.9604	0.0143	67.3768	0.8574
6	0.0175	0.0564	0.3099	0.9594	0.0125	76.4516	0.8856
7	-0.0264	0.0519	-0.5077	0.9741	0.0115	84.3974	0.9042
8	0.1402	0.0541	2.5891	0.9798	0.0120	81.4162	0.8977
9	0.1320	0.0529	2.4933	0.9627	0.0118	81.8067	0.8986
10	0.1134	0.0621	1.8267	0.9639	0.0138	69.8276	0.8659

Discussion. Betas range from 0.9594 to 1.3030. Decile 1 is the most sensitive to market movements. The market explains about 77.7%–90.4% of the variation in monthly portfolio excess returns, so the regressions have substantial explanatory power.

Using conventional standard errors at the two-sided 5% level, deciles 1 and 3 have significantly negative alphas, while deciles 8 and 9 have significantly positive alphas. These findings suggest that CAPM does not fully explain average returns for every decile.

3 Heteroskedasticity and serial correlation

Heteroskedasticity. After each OLS regression, we run `estat hettest`, the Breusch–Pagan/Cook–Weisberg test using fitted values. The null is constant error variance. This default test assumes normal errors and uses a $\chi^2(1)$ reference distribution.

Serial correlation. We run `estat bgodfrey, lags(12)`, the Breusch–Godfrey test. The null is no residual serial correlation at lags 1 through 12 jointly. Twelve monthly lags cover one year. The test uses a $\chi^2(12)$ reference distribution; Stata fills unavailable initial residual lags with zero.

Table 2. OLS residual diagnostics

Decile	Heteroskedasticity (BP)		Serial correlation (BG)	
	$\chi^2(1)$	<i>p</i> -value	$\chi^2(12)$	<i>p</i> -value
1	0.0492	0.8245	38.2389	0.0001
2	3.8861	0.0487	33.4207	0.0008
3	0.4444	0.5050	13.4441	0.3376
4	4.0636	0.0438	10.3261	0.5874
5	2.4120	0.1204	11.4735	0.4888
6	0.9608	0.3270	20.9560	0.0510
7	1.2558	0.2624	30.1811	0.0026
8	1.3165	0.2512	22.7785	0.0297
9	0.3125	0.5761	21.7645	0.0402
10	4.5681	0.0326	27.1589	0.0073

Results at 5%. The heteroskedasticity test rejects constant variance for deciles 2, 4 and 10. The serial-correlation test rejects for deciles 1, 2, 7, 8, 9 and 10. Decile 6 is just above the threshold ($p = 0.0510$).

The BG test is a conventional diagnostic and is not itself robust to heteroskedasticity. For final inference, we use Newey–West standard errors for every portfolio, regardless of the diagnostic results.

4 CAPM and hypothesis tests with Newey–West standard errors

We run `newey excessd market, lag(12)` for each decile. Newey–West HAC standard errors allow heteroskedasticity and account for serial correlation through 12 monthly lags using Bartlett weights. Coefficients and R^2 remain the same as OLS; standard errors and test statistics change.

Table 3. CAPM estimates with Newey–West standard errors

Decile	α	SE(α)	t_α	β	SE(β)	t_β	R^2
1	-0.3414	0.1357	-2.5153	1.3030	0.0452	28.8043	0.7766
2	-0.1437	0.0808	-1.7772	1.0980	0.0307	35.7948	0.8313
3	-0.1475	0.0584	-2.5251	1.0298	0.0240	42.9491	0.8696
4	-0.0196	0.0665	-0.2953	0.9706	0.0227	42.7606	0.8720
5	0.0904	0.0620	1.4577	0.9604	0.0229	41.9145	0.8574
6	0.0175	0.0569	0.3076	0.9594	0.0176	54.4487	0.8856
7	-0.0264	0.0510	-0.5167	0.9741	0.0154	63.3696	0.9042
8	0.1402	0.0568	2.4658	0.9798	0.0178	55.0466	0.8977
9	0.1320	0.0575	2.2971	0.9627	0.0169	56.8863	0.8986
10	0.1134	0.0711	1.5957	0.9639	0.0266	36.2165	0.8659

We test $H_0 : \beta_d = 1$ against $H_1 : \beta_d \neq 1$ using $t = (\hat{\beta}_d - 1)/SE_{HAC}(\hat{\beta}_d)$ and the t_{755} distribution. At 5%, a significantly above-one beta is cyclical and a significantly below-one beta is defensive.

Table 4. Beta-equals-one tests with Newey–West standard errors

Decile	$t(\beta = 1)$	p -value	Conclusion at 5%
1	6.6984	< 0.0001	Cyclical
2	3.1944	0.0015	Cyclical
3	1.2429	0.2143	Not distinguishable from 1
4	-1.2974	0.1949	Not distinguishable from 1
5	-1.7266	0.0846	Not distinguishable from 1
6	-2.3064	0.0214	Defensive
7	-1.6864	0.0921	Not distinguishable from 1
8	-1.1363	0.2562	Not distinguishable from 1
9	-2.2017	0.0280	Defensive
10	-1.3553	0.1757	Not distinguishable from 1

Final results. Deciles 1–2 are cyclical; deciles 6 and 9 are defensive. The other betas are not significantly different from one. Compared with conventional OLS tests, deciles 3, 4, 5, 7 and 10 lose significance. The alpha conclusions remain unchanged: deciles 1 and 3 have significantly negative alphas, and deciles 8 and 9 significantly positive alphas. All p -values are unadjusted for multiple testing.

APPENDIX A

Stata code

```

1  version 19.0
2  clear all
3  set more off
4
5  * 1. Downloaded from Kenneth French's Data Library on 15 September 2026.
6  * Raw ZIPs and CSVs are preserved in data/raw for reproducibility.
7  * Original URLs:
8  * https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/Portfolios_Formed_on_OP_CSV.zip
9  * https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/ftp/F-F_Research_Data_Factors_CSV.zip
10 * Row ranges select only the monthly blocks in this frozen vintage.
11 * OP columns 10-19 are the ten value-weighted decile portfolios.
12 import delimited using "data/raw/Portfolios_Formed_on_OP.csv", ///
13     rowrange(26:782) varnames(nonames) asdouble clear
14 rename v1 yyyymm
15 forvalues d = 1/10 {
16     local c = 'd' + 9
17     rename v'c' ret'd'
18 }
19 keep yyyymm ret1-ret10
20 isid yyyymm
21 tempfile portfolios
22 save 'portfolios'
23
24 * Market excess return is column 2; RF is column 5.
25 * Mkt-RF is already an excess return: do not subtract RF again.
26 import delimited using "data/raw/F-F_Research_Data_Factors.csv", ///
27     rowrange(6:1206) varnames(nonames) asdouble clear
28 rename v1 yyyymm
29 rename v2 market
30 rename v5 rf
31 keep yyyymm market rf
32 isid yyyymm
33 merge 1:1 yyyymm using 'portfolios'
34 assert _merge != 2
35 keep if _merge == 3
36 drop _merge
37 generate month = ym(floor(yyyymm/100), mod(yyyymm,100))
38 format month %tm
39 sort month
40 tsset month
41 assert month == month[_n-1] + 1 if _n > 1
42 assert _N == 757
43 assert month[1] == ym(1963,7) & month[_N] == ym(2026,7)

```

APPENDIX A / CONTINUED

Stata code

```

44
45 * 2. Returns and RF are in percent per month; retain these units.
46 foreach v of varlist ret1-ret10 market rf {
47     replace 'v' = . if inlist('v', -99.99, -999)
48     assert !missing('v')
49 }
50 forvalues d = 1/10 {
51     generate double excess'd' = ret'd' - rf
52     label variable excess'd' "OP decile 'd': excess return (%)"
53 }
54 label variable month "Month"
55 label variable rf "One-month risk-free return (%)"
56 label data "Value-weighted OP deciles: July 1963-July 2026"
57 save "output/op_excess_returns.dta", replace
58 export delimited using "output/op_excess_returns.csv", replace
59
60 * 3. CAPM: excess_d = alpha_d + beta_d * market + error_d.
61 * Default OLS standard errors; coefficient t-statistics test zero.
62 tempfile statistics hacstats
63 tempname results hacresults bgstat bgpval
64 postfile 'results' byte decile int n double alpha se_alpha t_alpha ///
65     beta se_beta t_beta r2 t_beta1 p_beta1 bp_chi2 bp_p bg_chi2 bg_p ///
66     using 'statistics', replace
67 postfile 'hacresults' byte decile int n double alpha se_alpha t_alpha ///
68     beta se_beta t_beta r2 t_beta1 p_beta1 using 'hacstats', replace
69 forvalues d = 1/10 {
70     quietly regress excess'd' market
71     * Breusch-Pagan test: H0 is constant error variance.
72     quietly estat hettest
73     local bp = r(chi2)
74     local bpp = r(p)
75     * Joint Breusch-Godfrey test of residual lags 1 through 12.
76     quietly estat bgodfrey, lags(12)
77     matrix 'bgstat' = r(chi2)
78     matrix 'bgpval' = r(p)
79     local fit = e(r2)
80     local t1 = (_b[market] - 1) / _se[market]
81     local p1 = 2*ttail(e(df_r), abs('t1'))
82     post 'results' ('d') (e(N)) (_b[_cons]) (_se[_cons]) ///
83         (_b[_cons]/_se[_cons]) (_b[market]) (_se[market]) ///
84         (_b[market]/_se[market]) (e(r2)) ('t1') ('p1') ('bp') ('bpp') ///
85         ('bgstat'[1,1]) ('bgpval'[1,1])
86     * 4. Two-sided test of beta = 1; equivalent to t1 squared.
87     test market = 1
88     assert abs(r(p) - 'p1') < 1e-8

```

APPENDIX A / CONTINUED

Stata code

```

89      * Newey-West HAC: heteroskedasticity and 12 monthly serial lags.
90      quietly newey excess'd' market, lag(12)
91      local t1 = (_b[market] - 1) / _se[market]
92      local p1 = 2*ttail(e(df_r), abs('t1'))
93      post 'hacresults' ('d') (e(N)) (_b[_cons]) (_se[_cons]) ///
94          (_b[_cons]/_se[_cons]) (_b[market]) (_se[market]) ///
95          (_b[market]/_se[market]) ('fit') ('t1') ('p1')
96      test market = 1
97      assert abs(r(p) - 'p1') < 1e-8
98  }
99  postclose 'hacresults'
100 postclose 'results'
101 use 'statistics', clear
102 generate str16 classification = "Not distinguish."
103 replace classification = "Cyclical" if p_betal < .05 & beta > 1
104 replace classification = "Defensive" if p_betal < .05 & beta < 1
105 format alpha-r2 t_betal %9.4f
106 format p_betal %10.7f
107 list, noobs separator(0)
108 save "output/capm_results.dta", replace
109 export delimited using "output/capm_results.csv", replace
110
111 use 'hacstats', clear
112 generate str16 classification = "Not distinguish."
113 replace classification = "Cyclical" if p_betal < .05 & beta > 1
114 replace classification = "Defensive" if p_betal < .05 & beta < 1
115 format alpha-r2 t_betal %9.4f
116 format p_betal %10.7f
117 list, noobs separator(0)
118 save "output/capm_hac_results.dta", replace
119 export delimited using "output/capm_hac_results.csv", replace

```